

A PROJECT TITLE
ON

PLANT DISEASE DETECTION USING CNN AND TRANSFER LEARNING
project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

In

A.Y.: 2026-27

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DEPARTMENT OF CSE

Name of Title: PLANT DISEASE DETECTION USING CNN AND TRANSFER LEARNING

1. Abstract:

Agriculture plays a significant role in the economy, and plant diseases can greatly affect crop quality and yield. Early and accurate detection of plant diseases is essential for reducing crop losses and improving agricultural productivity. Traditional methods of identifying plant diseases require expert knowledge and are often time-consuming. This project presents an intelligent plant disease detection system using Convolutional Neural Networks (CNN) and Transfer Learning to automate the disease identification process. The proposed system uses images of plant leaves as input. Image preprocessing techniques such as resizing and normalization are applied before feeding the images into a pre-trained deep learning model. Transfer Learning is employed using a pre-trained CNN model such as MobileNetV2, ResNet50, or EfficientNet to improve classification accuracy while reducing training time. The model extracts important features from leaf images and classifies them into healthy or diseased categories. The dataset consists of labeled images of various plant diseases collected from publicly available agricultural datasets. During training, the model learns disease-specific patterns and features. The trained model is evaluated using performance metrics such as accuracy, precision, recall, and F1-score. Experimental results demonstrate that Transfer Learning significantly improves disease classification accuracy compared to training a CNN model from scratch. The developed system can assist farmers and agricultural experts by providing quick, reliable, and cost-effective disease diagnosis. Early disease detection enables timely treatment, minimizes crop losses, and increases overall productivity. The proposed solution is scalable and can be integrated into mobile or web applications for real-time field usage. This project highlights the potential of Artificial Intelligence and Deep Learning in promoting smart agriculture and sustainable farming practices.

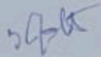
2. INTRODUCTION:

Agriculture is one of the most important sectors contributing to food production and economic development. Plant diseases pose a serious threat to agricultural productivity by reducing crop quality and yield. Identifying diseases at an early stage is essential for preventing their spread and minimizing financial losses. Recent advancements in Artificial Intelligence and Deep Learning have made automated plant disease detection possible. Convolutional Neural Networks (CNNs) have shown excellent performance in image classification tasks due to their ability to learn complex visual features. Transfer Learning further improves the efficiency of CNN models by utilizing knowledge gained from large pre-trained datasets, reducing both training time and computational cost. This project aims to develop an automated plant disease detection system using CNN and Transfer Learning. The proposed model analyzes leaf images and accurately identifies different plant diseases. Such a system can support farmers in making timely decisions, reducing dependency on manual inspection, and improving agricultural productivity. In addition, the proposed plant disease detection system provides an efficient and reliable approach for identifying diseases from plant leaf images. In traditional agriculture, farmers often depend on visual observation and expert advice to identify diseases, which may take considerable time and can sometimes lead to incorrect identification, especially when different diseases show similar symptoms. The proposed system overcomes these limitations by using Artificial Intelligence and Deep Learning techniques to automatically analyze the visual patterns present in leaf images. The CNN model is capable of learning important features such as leaf color, texture, shape, spots, lesions, and other disease-related patterns from the training images. Transfer Learning makes the system more effective by using the knowledge of an already trained deep learning model and adapting it to the plant disease classification task. This reduces the amount of training required and improves the overall efficiency of the system. The model can classify the input leaf image into appropriate healthy or diseased categories based on the learned features. Such an automated

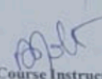
3. SOFTWARE REQUIREMENTS:

Operating System: Windows 10/11, Ubuntu, or macOS
Programming Language: Python 3.10 or above
IDE: Jupyter Notebook / Visual Studio Code / PyCharm
Deep Learning Framework: TensorFlow 2.x
Machine Learning Library: Keras
Libraries: NumPy, Pandas, Matplotlib, OpenCV, Scikit-learn
Dataset: PlantVillage Dataset
Hardware: Minimum 8 GB RAM, Intel i5/Ryzen 5 Processor, GPU (Optional but Recommended)
Version Control: Git & GitHub

This content is suitable for a Machine Learning B.Tech project report and can be directly copied into your template.



Signature of the Guide



Course Instructor